A resistive-strip Micromegas TPC for the X17 search at n_TOF: design, track reconstruction and performance

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Session: Gas detectors

The X17 search at the CERN n_TOF neutron facility requires tracking e^+e^- pairs from neutron capture on ^{3}He. To test the anomaly, their opening angle must be measured with as little material as possible. We developed a resistive-strip bulk Micromegas that works as a compact TPC: one 30 mm drift gap gives a three-dimensional track segment. Its $40 \times 40\,\mathrm{cm}^2$ area is read out through a pad matrix with a 0.78 mm pitch, capacitively coupled to two orthogonal strip views.

The resistive layer presents unique challenges for TPC operation. Charge spreads to the neighbouring strips with an RC delay, and this distorts the arrival time measured on each strip: angles taken from those times read about 4° too steep. We developed a novel reconstruction that works at the waveform level: it fits the full signal of every strip in a cluster at once, with a model of the charge sharing, and recovers the true track.

The chamber reaches an angular resolution of 1.2°, 93 % efficiency and a spatial resolution of 180 µm. At n_TOF, the reconstructed tracks image the ^{3}He target capsule, demonstrating the detector's efficacy as a TPC tracker.